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Beyond oak regeneration: modelling mesophytic sapling density drivers along topographic, edaphic, and stand-structural gradients in mature oak-dominated forests of Appalachian Ohio

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Abstract

Shade-tolerant mesophytic tree species tend to dominate the understories of present-day oak-hickory and mixed-hardwood forests in the eastern United States. We quantified the sapling density associations with abiotic and biotic variables for three important mesophytic species: red maple (*Acer rubrum*), sugar maple (*Acer saccharum*), and American beech (*Fagus grandifolia*) in southeastern Ohio. In this study, we sampled 165 permanent plots in oak-dominated, topographically diverse, mature (>90 years old), second-growth forests, following 21-25 year timespan (1993-95 and 2016-18) between samples on the Athens and Marietta Units of the Wayne National Forest. Our models showed that sugar maple was strongly associated with high pH soils, and red maple with low pH soils. Additionally, red maple was associated with upper slope positions and older stands, while American beech was associated with lower slopes, northeasterly aspects, and northeast-facing upper slopes. Basal area of competing species, solum depth, and management unit were not significantly related to sapling density for our focal

species. American beech sapling density doubled between the two sampling periods, while both maple species' densities declined by half. Our results will help scientists and managers by providing insight into potential future composition of currently oak-dominated forests, in areas without active management intervention.

Keywords: Maple, beech, saplings, oak forests, mesophication

Introduction

Shade-tolerant mesophytic tree species are dominating the regeneration layer of many oak-hickory (*Quercus* spp. and *Carya* spp.) and mixed-hardwood forests across the eastern U.S. (Nowacki and Abrams 2008; McEwan et al. 2011; Pederson et al. 2015), while the density of oak in the canopy is decreasing (Fei et al. 2011; Knott et al. 2019). Compared to oak regeneration, ecological relationships among mesophytic species have received relatively little attention in the literature of oak forest management (Lorimer et al. 1994; Larsen and Johnson 1998; Brose et al. 2007; Povak et al. 2008; Miller et al. 2017; Graboski et al. 2020; Izbicki et al. 2020). Oak restoration is clearly a critical ecological issue that deserves the detailed attention it has received (Kirkpatrick and Pekins 2002; Rodewald and Abrams 2002; McShea et al. 2007; Tallamy and Shropshire 2009; Wood et al. 2012; Iverson et al. 2019a; Radcliffe 2019; Sierzega and Eichholz 2019; Zarri et al. 2020). However, the knowledge gap about mesophytic species is problematic in places like southeastern Ohio, where multiple mesophytic species are common in small to medium size classes and where a “hands-off” approach has been a dominant management strategy on many landownership types (Widmann 2016). In this region, red maple (*Acer rubrum*), sugar maple (*Acer saccharum*), and American beech (*Fagus grandifolia*) (hereafter,

“beech) are common mesophytic species in regeneration layers of mature hardwoods forests (Palus et al. 2018; Dyer and Hutchinson 2019). Each of these species provides varying amounts of different ecosystem services and faces unique threats. For example, sugar maple is the most economically valuable of the three species, for lumber (Dey et al. 2017) and for syrup (Matthews and Iverson 2017). Beech provides hard mast crops if healthy, and thus an increase in beech would likely be beneficial to wildlife that also consume acorns and hickory nuts (Nixon et al. 1968; Gysel 1971; Jakubas et al. 2004; Tissier et al. 2020). Beech faces serious threats from beech bark disease (Cale et al. 2017) and beech leaf disease (Ewing et al. 2018), while both maple species are potentially threatened by Asian long-horned beetle (*Anoplophora glabripennis*) (Dodds and Orwig 2011) and spotted lanternfly (*Lycorma delicatula*) (Dara et al. 2015). Climate change impacts over the next century are expected to influence each species in unique ways, with red maple potentially adapting better than sugar maple or beech (Butler et al. 2015; Iverson et al. 2019b). Forest managers must understand species-specific patterns and drivers of red maple, sugar maple, and beech regeneration, if they wish to anticipate which ecosystem services unmanaged forests may provide in the future.

Influence of abiotic factors

Topographic variables such as slope position and slope aspect are important influencers of regeneration patterns for our focal species. In mature forests of the Central Hardwood region, red maple saplings and small trees are generally most common on upper slopes and other drier landscape positions (Jose and Gillespie 1997; Arthur et al. 2015; Chapman and McEwan 2016). Conversely, sugar maple grows best and is often most common on lower slopes (Jose and Gillespie 1997; St Clair and Lynch 2005; Cleavitt et al. 2014), north slopes (McCarthy et al. 2001; Chapman and McEwan 2016) and mesic landscape positions (Arthur et al. 2015; Chapman

and McEwan 2016). In Central Hardwood forests, beech saplings and small trees are most common on mesic landscape positions, although they can be relatively ubiquitous across the landscape (Jose and Gillespie 1997; Palus et al. 2018). Root suckering can allow beech to regenerate in conditions that would otherwise be unsuitable (Held 1983), including drier and more upper slopes (Takahashi et al. 2010).

Researchers have frequently shown that more acidic soil is unfavorable to sugar maple (Bal et al. 2015). This is often attributed to the decreasing availability of calcium and magnesium and the increasing availability of aluminum and manganese with decreasing pH, although many other nutrients important for sugar maple become less available with decreasing pH (Houle et al. 2007; St.Clair et al. 2008; Bal et al. 2015). Relative to sugar maple, red maple (McDonald et al. 2002; St Clair and Lynch 2005; Zarfos et al. 2019) and beech (Long et al. 1997; Bigelow and Canham 2002; Duchesne et al. 2002) both tend to be more common on more acidic soils. Most of this evidence, however, comes from northern hardwoods ecosystems of the northeastern U.S. and southeastern Canada (but see: Nigh et al. 1985), where soil tends to be more acidic (Brady and Weil 2010). It is unclear whether sugar maple is being affected by soil pH in southeastern Ohio to the same extent that it is in northern hardwoods forests.

In southeastern Ohio, soil depth is a likely driver of canopy composition (Hix and Pearcy 1997) and forest productivity (Carmean 1965). The solum (A and B horizon of the soil profile) is often less than a meter deep in our study sites, and soil depth could be an important driver of sapling dynamics. We hypothesized that the relatively stress tolerant red maple (Abrams 1998) would be more competitive on thinner soils than the more mesophytic species sugar maple and beech.

Influence of stand-level biotic factors

Canopy cover is likely to affect density of our focal species; several studies have shown that beech has an advantage in growth and survival in closed-canopy situations, while sugar maple can have a growth advantage in individual-tree canopy gaps (Canham 1988; Canham et al. 1994; Kobe et al. 1995). These observations have also held true at the stand scale (Poulson and Platt 1996; Nolet et al. 2008). Compared to beech and sugar maple, red maple is less tolerant (Burns and Honkala 1990; Canham et al. 1994). However, it can withstand different light levels, surviving periods of suppression in the understory (Tift and Fajvan 1999; Hart et al. 2012), and competing effectively in relatively open stands (Rankin and Pickett 1989; Fei and Steiner 2009).

Stand age is also likely an important factor for saplings of our focal species. A chronosequence study in southeastern Ohio found a pattern of increasing beech sapling abundance with age, greater red maple sapling abundance in the oldest stands (greater than 150 years), and higher sugar maple abundance in somewhat younger stands (90-130 years) (Goebel and Hix 1996). Increasing stand age in undisturbed forests is linked to increasing basal area and competition (Goebel and Hix 1996; Woods 2000), which may be expected to favor beech over sugar maple and red maple.

Relative Abundance Patterns

A recent pattern of increasing beech abundance and decreasing sugar maple abundance in the regeneration layer over time has been noted in several studies of northern hardwoods (Duchesne and Ouimet 2009; Gravel et al. 2011). Possible explanations include beech's extreme tolerance giving it a competitive advantage over maple as many forests advance in successional development (Canham et al. 1994), increasing whitetail-deer (*Odocoileus virginianus*) browsing sugar maple more than beech (Lesser et al. 2019), and increasing soil acidity from beech leaf litter negatively affecting sugar maple (Hane et al. 2003). However, the most common

explanation is decreasing availability of base cations and increasing soil acidity caused by atmospheric deposition, which can incite sugar maple decline and result in gaps for possible beech expansion (Duchesne and Ouimet 2009; Gravel et al. 2011). Red maple has also been increasing across most the Central Hardwoods region and beyond during the past several decades (Lorimer 1984; Fei and Steiner 2007; Zhang et al. 2015), perhaps due to red maple gradually altering conditions on the forest floor to favor itself (Kreye et al. 2013; Alexander and Arthur 2014), acid rain favoring red maple over other tolerant species (Bigelow and Canham 2002), and/or red maple's ecological flexibility along multiple resource gradients (Abrams 1998).

Trends in Relationships with Environmental Variables over Time

Topographic variables may be exerting less influence over mesophytic tree species than in former decades and centuries (Hanberry et al. 2012), potentially due to fire suppression (Nowacki and Abrams 2008) and positive feedback loops between mesophytic species and forest floor conditions (Kreye et al. 2013, 2018; Alexander and Arthur 2014). For example, *Acer* species were prevalent on drier topographic positions in the modern-era when compared to pre-settlement data in southern Missouri (Hanberry et al. 2012) and southern Ohio (Dyer and Hutchinson 2019). Additionally, Chapman and McCarthy (2018) documented beech regeneration expanding to upper slope positions, and oak and hickory regeneration contracting away from lower slope positions, over four sampling periods during a 31-year timespan in an old-growth stand in eastern Kentucky.

Objectives and Hypotheses

Our study sought to quantify associations of biotic and abiotic drivers of sapling density for the three most common mesophytic species in southeastern Ohio: red maple, sugar maple, and

beechn. We sampled two disjunct units of the Wayne National Forest (hereafter the Wayne), in two time periods: the 1990s and 2010s. Using these data, we explored three hypotheses:

Hypothesis 1: Topographic, edaphic, and stand-structural factors are strongly associated with sapling density. Sugar maple and American beech saplings will tend to be more abundant on more mesic topographic positions, deeper soil, and more closed canopy positions. Sugar maple will be more abundant on soils with higher pH values, and beech on soils with lower pH values. Red maple will be more abundant on more xeric topographic positions, shallower soil, lower pH soil, and in more open canopy positions.

Hypothesis 2: The association of topographic variables with sapling density will become weaker over time, due to mesophytic species creating positive feedback loops that increase the forest floor and soil moisture levels over time.

Hypothesis 3: American beech and red maple sapling densities will increase from the first to the second sampling period while sugar maple sapling density will decrease.

Methods

Study Area

Our study area includes two administrative units of the Wayne National Forest in southeastern Ohio, the Athens Unit and the Marietta Unit (Figure 1). The units are spatially disjunct and separated by a minimum of 37 kilometers. Southeastern Ohio (AKA “Appalachian Ohio”) is unglaciated, strongly dissected by steep valleys, heavily forested, and has a long history of intensive and extensive land use, settlement, and land abandonment by Native

158 American groups and European descendants (Hutchinson et al. 2003; Iverson et al. 2019a; Kern
159 and Wilson 2014; Whitney 1994).

160 **Climate**

161 Southeastern Ohio has a continental climate with cold winters and warm, humid
162 summers. It receives precipitation (mostly rainfall) relatively evenly throughout the year,
163 although rainfall is slightly lower in fall and winter than in spring and summer (NOAA 2020).
164 The Marietta Unit is likely slightly moister than the Athens Unit; the average annual rainfall for
165 the Marietta Wastewater Treatment Plant weather station was 108 centimeters (cm) from 1981-
166 2010, 8 cm more than the 100 cm average for the Athens Ohio University weather station
167 (NOAA 2020). Additionally, the temperature for the Marietta Station was slightly higher than
168 that of the Athens Station for the same period (Marietta average mean temp 11.7°C, average min
169 temp 5.6°C, average max temp 17.7°C; Athens average mean temp 11.0°C, average min
170 temperature 5.2°C, average max temperature 16.8°C, on an annual basis) (NOAA 2020).

171 **Bedrock and Soil**

172 The Athens Unit and Marietta Unit are within the unglaciated Allegheny Plateau
173 Province (Lessig et al. 1977; Lucht et al. 1985). The Athens Unit is primarily located on slightly
174 older bedrock, from the Pennsylvanian Era of 302-318 million years ago, formed from multiple
175 cycles of advance and retreat of prehistoric seas and swamp forests. Most of the bedrock of the
176 Marietta Unit is from the Permian Era of 298-302 million years ago, formed from sediment
177 accumulation in a prehistoric coastal swamps, with some Pennsylvanian bedrock in modern
178 valleys (Kern and Wilson 2014; Ohio Division of Geological Survey 2017). Common rock types
179 of both units include sandstone, shale, siltstone, limestone, and coal (Lessig et al. 1977; Lucht et
180 al. 1985). Most soils in the region are of the order Alfisol, suborder Udalf, and small areas in

181 each unit are the soil order Ultisol, suborder Udult (Brady and Weil 2010). The udic designation
182 of both suborders indicates a relatively wet moisture regime, adequate for plant growth year
183 round (Brady and Weil 2010).

184 The physiography of the Athens Unit and Marietta Unit is greatly dissected with steeply
185 sloping valleys and level bottomlands and ridgetops. There are frequently two to three relatively
186 flat structural benches along the length of a typical hillslope. These benches are artifacts of the
187 multi-layered nature of the bedrock, and the differential erosion rates of the layers (Hix and
188 Percy 1997).

189 **Vegetation**

190 Southeastern Ohio is within the Central Hardwoods region, Appalachian Mountain
191 section of Fralish (2003), the mixed-mesophytic forest region of Braun (1950), and the
192 mesophytic forest region of Dyer (2006). It is characterized by a high diversity of primarily
193 deciduous tree species. The most common species by importance value in Dyer's mesophytic
194 forest region were red maple (11%) and white oak (*Quercus alba*) (5%), based on Forest
195 Inventory and Analysis (FIA) data (Dyer 2006, page 347). Presettlement land surveys of
196 southeastern Ohio indicated oaks comprised 61% of the witness trees, while red maple, sugar
197 maple, and beech combined made up 18% of the witness trees (Dyer and Hutchinson 2019, page
198 34). In southeastern Ohio forests studied by Hutchinson et al. (2003), these three mesic species
199 were mostly limited to bottomlands, and red maple in particular was uncommon. Recent studies
200 of the Athens Unit of the Wayne National Forest (Palus et al. 2018), and in other areas of
201 southeastern Ohio (Deines et al. 2016; Dyer and Hutchinson 2019) have found sugar maple and
202 red maple have much greater abundance and more widespread distribution than was indicated by
203 the presettlement land surveys. Forests in southeastern Ohio have relatively low browsing

pressure from whitetail deer (Apsley and McCarthy 2004), possibly because relatively unfragmented land ownership and high antlerless permit sales may increase harvest rates (Karns et al. 2016).

Study Design

Our study is a continuation of the Wayne National Forest Ecological Classification System (Wayne ECS) project, a long-term study for which a network of permanent plots was established on the Wayne National Forest in the early to mid-1990s (Hix and Chech 1993; Hix and Percy 1997). Our work contrasts with that of Palus et al. (2018), which analyzed changes in sapling and overstory composition on the Athens Unit of the Wayne in relation to ecological land types that were delineated using topographical variables. Our study used 83 of the same sapling plots from the Athens Unit. However, this study is novel in specifically modelling the relationships of sapling abundance with edaphic and stand-structural variables, as well as continuous topographic variables, with a focus on the most important mesophytic species.

We sampled 165 nested permanent plots from 30 stands distributed across the landscape of the study area, with a 21-25 year window between repeated samples (1993-95 and 2016-2018). Researchers originally established 128 permanent plots along 58 transects in the Marietta Unit in 1993 – 1995 (Hix and Percy 1997). We resampled 82 of these plots over the summers of 2017 and 2018; the remainder were either lost due to access issues or forest harvesting, or had a missing datapoint for one of our covariates, which disqualified them from the current study. Researchers also established 108 permanent plots along 51 transects in the Athens Unit in 1993-1995 (Hix and Chech 1993), and we resampled 100 of these plots in 2016 (Palus et al. 2018). We used 83 plots in our analysis of the Athens Unit, due to missing data points for covariates of 17 of the resampled Athens plots.

Hix and Chech (1993) originated the study to be representative of mature, relatively undisturbed second-growth forests (at least 70 years old at the original time of sampling, and at least 93 years old at time of first resampling, based on U.S. Forest Service Vegetation Management Information System data and visual inspection). The stands were originally chosen on the basis of five criteria: “(1) no obvious evidence of major anthropogenic disturbance within the past 40 years (e.g., no stumps), (2) no dominance of early successional tree species, (3) no evidence of recent fire, significant windthrow, or unexplainable mortality, (4) stand is ‘fully stocked’ relative to the upland Central Hardwoods stocking chart (Gingrich 1967), and (5) no extensive (multiple-tree) gaps in the canopy” (Hix and Percy 1997, page 1118-1119). Within a random subset of each stand that met these criteria, the original researchers established one or two transects running roughly through the center of the stand. Each transect contained two to five plots running parallel with elevational contours. Each plot was located between 30 and 50 meters from the next plot on the transect.

Vegetation Sampling

Each “plot” consisted of two circular nested subplots, of 100 m² and 500 m² sizes. Within the 100-m² subplot, we tallied by species each individual woody stem with a diameter at breast height (1.37 m; dbh) of 0.1-10.0 cm. We refer to these plants as the “sapling layer,” and the data includes tree and shrub species. Within the 500-m² subplot, we recorded the species and dbh to the nearest tenth-centimeter of each woody plant with a dbh of 10.1 cm or more, alive or dead. We refer to these plants as the “tree layer,” including those that are overtopped or intermediate. We visually estimated deer browsing pressure on a 1 to 5 scale, using SILVAH protocols (Knopp and Stout 2014).

Environmental Variables and Soils

The researchers that initially measured the plots recorded the environmental variables that we used as predictor variables in our models. These are slope percent, aspect, and percent distance to ridge. The latter was determined by mapping the length of the slope from ridgetop to bottomland, in a line that passed through each plot and was perpendicular to the slope at the point of the plot, and then dividing the distance of the plot from the ridge by the total length of the slope. One soil pit per transect was excavated and the following measurements were determined by sub-horizon: texture, pH, depth, and thickness (Hix and Chech 1993).

Data Analysis

To model focal species' sapling abundance, we used generalized linear mixed models (glmm) with negative binomial distribution assumptions. The negative binomial distribution is commonly used for count data, and we chose it over the Poisson distribution because of observed overdispersion of variance with our three focal species, better model fits with negative binomial assumptions, and better visual fits of our data with the negative binomial distribution. We used transect as a random effect, in order to control for potential pseudo-replication caused by grouping our plots in space (Zuur et al. 2010), especially in regards to the soil data, for which one pit was used per transect. We ran models using the "glmer.nb" function in the lme4 package (Bates et al. 2019) in R version 3.4.4.

For model evaluation, we selected a set of *a priori* variables and then compared the common model with each variable included for each unit and sampling period. Keeping the same set of variables in each final model afforded more direct comparison of effect sizes between different species and time periods, compared to common parameter reduction methods

(Burnham and Anderson 2002). We used seven continuous predictor variables spanning three distinct categories: landscape variables (transformed slope aspect, percent distance to ridge, and the interaction factor of the two), edaphic variables (solum depth, acidity of B horizon), and stand-structural variables (basal area of the plot, stand age). We also included management unit of the Wayne as a categorical predictor variable, to test for differences between units not accounted for by other variables. Variable justification, relevant studies, our hypothesized relationships, and further explanation of the units chosen are contained in Table 1, and summary statistics of predictive variables are shown in Table 2.

All variables were screened for covariance with a threshold of 0.7. We also checked multicollinearity using variance inflation factors (VIF), using the “vif” function in package car (Fox et al. 2020). VIF values of predictor variables were well below a conservative threshold of three (Zuur et al. 2010) when the interaction term of transformed aspect to percent distance to ridge was not included ($VIF < 1.2$ for all predictors). However, transformed aspect and the interaction term rose above this threshold when the interaction factor was included (VIF 4.8 for transformed aspect, VIF 6.0 for interaction term). Thus, topographic results need to be interpreted in the context of all three variables we used and previous support for the interaction of these variables in the region of our study area (Adams et al. 2019; Iverson et al. 2019a).

We chose not to transform predictor variables despite some showing right skewness, because glmms do not assume predictor variable normality (Zuur et al. 2009) and we wanted to maintain clear interpretability of model outputs. We measured individual covariate significance with partial coefficient significance from model outputs with a p value threshold of 0.05. The overall predictive ability of each model was assessed using the trigamma method of predicting marginal and conditional r^2 value, which is a preferred method for a glmm with a log link

(Barton 2018). Marginal r^2 values assess variation explained by fixed effects alone, and conditional r^2 values assess variation explained by fixed and random effects (Nakagawa and Schielzeth 2013). We calculated these using the function “r.squaredGLMM” in the MuMIn package (Barton 2018).

To test significance of changes in sapling density over time, we used a nonparametric Mann Whitney U Test on plot level sapling counts. This accounted for the right-skewed nature of our count data and the high number of zero counts (McKnight and Najab 2010).

We calculated importance value of all canopy tree species and the relative density of all sapling species. We defined canopy-layer importance values as relative basal area + relative dominance/2, for all trees greater than 10 cm dbh. Since we didn’t measure the diameter of trees in our sapling layer, we calculated relative density for all saplings, including any woody species that met our height and diameter definitions for saplings.

Results

Sapling Abundance Trends and Models

Topographic variables showed several significant associations with density (Table 3), including percent distance to ridge for both red maple models (-), percent distance to ridge, transformed aspect, and the interaction factor of transformed aspect and percent distance to ridge for the beech 2010s model (+). Of the soil variables, pH of the B horizon exhibited significant associations with density for red maple models (-) and sugar maple models (+), while depth of solum did not display any significant associations. The only significant interaction for stand-structural variables was stand age for red maple 2010s (+). Wayne management unit did not

show significant interactions for any of the models. Table 3 shows summary statistics for all continuous covariates in our models.

The predictive ability of the models, as measured by conditional r^2 values, were highest for red maple (1990s: 0.68, 2010s: 0.82), followed by beech (1990s: 0.68, 2010s: 0.55) and sugar maple (1990s: 0.43; 2010s: 0.33). Coefficient estimates of transformed slope aspect and percent distance to ridge were closer to zero in the sugar maple model for the more recent sampling period. For the same two landscape coefficients, estimates moved further from 0 in the 2010s for the beech and red maple models (Table 4).

Among the three focal species, there were large changes in abundance over time ($p < 0.01$ for all three species). In the 1990s, sugar maple had the highest sapling density (463 saplings per hectare [sph] Athens, 532 sph Marietta), followed by red maple (320 sph Athens, 467 sph Marietta) and beech (144 sph Athens, 237 sph Marietta). In the 2010s, beech had the highest sapling density (457 sph Athens, 304 sph Marietta), followed by sugar maple (267 sph Athens, 233 sph Marietta) and red maple (256 sph Athens, 154 sph Marietta) (Table 5). Our three focal species combined decreased in density by about 23% between sampling periods. Overall, they comprised about 43% of relative density for all individual stems of woody species with a dbh between 0.1 and 10 cm, including all sampling periods and management units. They were the only species with relative density values above 10%. Each oak and hickory species had an average relative sapling density less than 1% (values in Table 5 are aggregated by genus for *Quercus* and *Carya*). Relative density of tree saplings may be interpreted as conservative, because subcanopy tree and shrub species were included in the analysis, and each woody stem was counted, whether or not it was clumped with conspecifics. Flowering dogwood (*Cornus florida*) had comparable densities to our focal species in the 1990s, but decreased to about 7% of

its former density by the 2010s. Meanwhile, spicebush (*Lindera benzoin*) nearly doubled in density between sample periods. Total sapling density increased slightly over time in the Athens (2,348 sph 1990s, 2400 sph 2010s, while it decreased sharply in the Marietta (2,693 sph 1990s, 1471 sph 2010s) (Table 5).

Overstory Trends and Deer Browse

Tree layer analysis showed that white oak was by far the most dominant species in terms of basal area (6.2 m²/ha 1990s, 7.5 m²/ha 2010s) and sugar maple had the highest density (63 trees per hectare [tpa] 1990s, 70 tpa 2010s), for both sampling periods. White oak had the highest importance value in both sampling periods (22.0 1990s, 20.0 2010s), but sugar maple increased in importance between the two sample periods (15.4 1990s, 17.6 2010s). From the 1990s through the 2010s, oaks exhibited a general trend of decreasing density and importance value but increasing basal area. Hickories followed the same trend, but the magnitude of change was small, partially due to their relatively low initial importance. Beech and both maple species increased in tree density and tree basal area with time (Table 6). Deer browse ratings showed low to moderate browsing pressure, never exceeding three on a scale of one to five.

Discussion

Ecological and management context

This study gives insight into potential consequences of foregoing management and suppressing disturbance in mature oak-dominated forests of the Central Hardwoods region. It is novel in focusing exclusively on patterns of mesophytic tree regeneration in an oak-dominated study area. Our three study species vary widely in economic value, value to wildlife, likely response to climate change, and threats posed by invasive insects and diseases. Therefore,

360 knowledge of where on the landscape each of these species are likely to dominate future oak-
361 dominated forests is critical to predicting future patterns of biodiversity and ecosystem service
362 provision. Although our focal species represent a small proportion of the total number of species
363 in our study area, they are by far the most dominant tree species in our sapling layer since the
364 early 1990s, and are common across the ecoregion (Iverson et al. 2019a) They are also known to
365 exert a high influence on woody plant community composition, because of their role as strong
366 light competitors that cast dense shade (Lorimer 1984; Burns and Honkala 1990; Runkle 1990;
367 Poulson and Platt 1996).

368 Our work should help managers when prioritizing areas for oak restoration. In
369 southeastern Ohio, identifying landscape areas where oak restoration is ecologically feasible has
370 been a major focus of recent research (Iverson et al. 2018; Iverson et al. 2019a), an important
371 application of recent advances in remote sensing technology (Adams et al. 2019, 2020), and a
372 major goal to be addressed in the plan revision process for the Wayne National Forest (USDA
373 Forest Service 2020b). We argue that the forest community types likely to result from foregoing
374 management should be another important consideration when choosing where to devote limited
375 oak restoration resources. For example, managers prioritizing economic value may choose to
376 retain sites likely to be sugar maple dominated in the future, while restoring oak where red maple
377 or beech are more likely to dominate in the future. Within the driest topographic positions
378 regionally identified as having the highest chance of success for oak regeneration (Iverson et al.
379 2018), our study showed that red maple is likely to be the most frequent species in the sapling
380 layer, with sugar maple likely to be common on some ridgetops. Beech appears not to be
381 regenerating in large numbers on ridgetops or upper southwesterly slopes, despite its general
382 increase in sapling importance.

383 **Relationships with environmental variables**

384 More generally, the importance of topographic and edaphic variables in our models is
385 consistent with our Hypotheses 1 and with a considerable body of work on topographically
386 complex areas of Central Hardwoods forests (e.g. Jose and Gillespie 1997; Olson et al. 2014;
387 Iverson et al. 2018; Adams et al. 2020). However the lack of a strong relationship of sugar
388 maple with slope position was counter to Hypothesis 1, because it contradicts results with
389 saplings and small trees in mature forests of Indiana (Jose and Gillespie 1997), Kentucky (Arthur
390 et al. 2015) and Missouri (Olson et al. 2014). This could indicate more favorable climate
391 conditions in our study area allowing sugar maple regeneration to survive in relatively harsh
392 environments, as is most likely the case when compared to Missouri, or more favorable soil
393 conditions. Alternatively, it could reflect a longer history of mesophytic species making
394 conditions more favorable for themselves on dry landscape positions (Hanberry et al. 2012; Dyer
395 and Hutchinson 2019), which would be consistent with our Hypothesis 2. We discuss some of
396 these possibilities below, but our results in context of prior studies suggest that the relative
397 influence of topography on sugar maple regeneration likely varies by region. In our study area,
398 sugar maple was significantly positively associated with more northeasterly slopes, indicating
399 topographic influences are important for regeneration of this species.

400 Notably, our models indicate that soil acidity is likely a strong mediator of competition
401 between red maple and sugar maple saplings, which dominate at opposite ends of the pH
402 spectrum with little overlap (Figure 2), consistent with Hypothesis 1. Red maple's relative
403 nutrient use efficiency likely allows it to grow with the lower nutrient availability generally
404 associated with more acidic soils (Collin et al. 2016). Our observational study design cannot
405 prove this as a causal mechanism. However, it is consistent with experimental research on

growth rates of maples related to calcium addition and/or pH (St Clair and Lynch 2005; Cleavitt et al. 2011; Marlow and Peart 2014), and other observational studies relating sapling growth (Bigelow and Canham 2002) and seedling abundance (Zarfes et al. 2019) to soil pH. To our knowledge, this is the first time soil acidity relationships with maple and beech saplings has been tested in southeastern Ohio; the majority of such studies have been conducted in northern hardwoods forests to the northeast of our study area (Bal et al. 2015), which tend to have had a higher proportion of Spodosols and higher overall soil acidity (Brady and Weil 2010).

Red maple was the only species to show significant relationships with our stand-structural variables of stand age and basal area; our red maple 2010s model showed a positive relationship with stand age. This may be due to mortality on dry sites; red oak (*Quercus rubra*) mortality is higher in older stands of the Marietta Unit (Radcliffe 2019), and scarlet oak (*Quercus coccinea*) mortality is high at relatively young stand ages (Goebel and Hix 1996; Radcliffe 2019). Mortality for either of these species, especially scarlet oak, could be creating gaps in drier topographic positions where red maple is best adapted to capitalize. Red maple is capable of capturing relatively small gaps (Tift and Fajvan 1999; Hart et al. 2012), and despite its lower tolerance than sugar maple or beech (Burns and Honkala 1990; Canham et al. 1994) it may be the most competitive late seral-competitor on dry sites, as evidenced by topographic relationships in our models (Figure 2).

Trends environmental variables over time

Our lack of evidence for Hypothesis 2 may be a result of our relatively short study period; previous studies testing this hypothesis have examined much larger timescales, comparing pre-European settlement General Land Survey data with modern Forest Inventory and

Analysis data (Hanberry et al. 2012; Dyer and Hutchinson 2019). Our marginal effects plots did show changes in slopes of predicted relationships of our environmental variables with our focal species over time (Figure 2), suggesting that the factors mediating regeneration competition are dynamic over time. However, confidence intervals overlapped between the two sample periods for each predictor variable within a given species (Radcliffe 2019). Longer time periods and additional study areas would be needed to determine whether these trends show consistent temporal patterns. Important questions of how relationships of species distributions and environmental variables change over time may be addressed by continuing permanent plot studies (Chapman and McEwan 2016). Such continuing efforts could lend further insight into observed patterns like that of sugar maple saplings' relative ubiquitousness across slope positions on northeasterly slopes, and questions about whether ridgetops and upper slopes will become less favorable for oak restoration with time (sensu: Nowacki and Abrams 2008).

Relative abundance patterns

Our study shows evidence that beech may be establishing a “second wave” of regeneration underneath a “first wave” of sugar maple and red maple. The observed dominance of maple in smaller diameter size classes for the tree layer (Radcliffe 2019) is likely resulting in more shading of the understory, which would be more favorable to beech than to either maple species (Runkle 1990; Canham et al. 1994). Several studies in northern hardwood forests have also noted patterns of increasing beech abundance over time in recent decades (Gravel et al. 2011). Deer browsing (Lesser et al. 2019) appears unlikely as an explanation given relatively low deer browsing pressure in our study area and southeastern Ohio (Karns et al. 2016), and we would expect stronger association of beech with soil acidity if either acid rain (Duchesne and Ouimet 2009; Gravel et al. 2011) or allelopathic qualities of beech litter (Hane et al. 2003) were

a main driver. Therefore, increasing overall basal area (Table 6) and associated decreases in light are a likely driver for increases in beech (Canham et al. 1994; Poulson and Platt 1996), despite our inability to show this association on a plot scale. The observed decrease in red maple saplings, contrary to our Hypothesis 3 and with regional trends in red maple abundance, would also be consistent with this explanation. Additionally, sugar maple is currently more dominant than beech (and red maple) in the tree layer (Radcliffe 2019), which could indicate a role for conspecific negative density dependence via Janzen-Connell effects (Forcier 1975; Woods 1979; Lambers et al. 2002; Johnson et al. 2014; Seiwa et al. 2019).

Beech saplings in northern hardwoods forests have higher survival and canopy-layer in-growth rates than sugar maple saplings (Gauthier et al. 2015), and if this pattern holds true in our study area, beech will likely continue to increase in importance relative to sugar maple. A more dominant presence of beech would potentially benefit mast-consuming animals in the long run (Gysel 1971; Jakubas et al. 2004; Tissier et al. 2020). Additionally, the topographic positions associated with beech saplings are different than those identified as most suitable for oak restoration (Iverson et al. 2018). Therefore, managers have a potential opportunity to manage for different mast producing species in different locations within the study area. Sugar maple and red maple, however, are often well-established in smaller size classes of trees (Radcliffe 2019). Sugar maple in particular is a long-lived species (Burns and Honkala 1990), so it will likely be important in the tree layer for the foreseeable future, especially considering its current importance in the tree layer.

However, trends in sapling dynamics are not necessarily deterministic of future canopy composition, because many disturbance agents could affect our study area. Arguably the most important example is beech bark disease, which may reach southeastern Ohio within the next

several decades (Cale et al. 2017). The resulting “aftermath forest” would probably still contain beech, but mostly in thick clumps of relatively small root suckers or advance sapling regeneration, rather than as mature trees (Cale et al. 2013; Giencke et al. 2014). Reducing the dominance of these beech clumps with management has proven difficult (Dracup and MacLean 2018; Elenitsky et al. 2020). While it might suppress new regeneration of species other than beech, beech bark disease invasion would likely favor established *Acer* saplings and small trees by releasing them from overstory competition (DiGregorio et al. 1999; Rumble et al. 2020), and possibly decrease soil acidity through reducing beech litter input (Arthur et al. 2017). The greatest collective effects on our focal species will likely be on northeasterly slopes, where beech and sugar maple saplings are abundant (Figure 2). This could have negative effects on mast-consuming wildlife (Gysel 1971; Jakubas et al. 2004; Tissier et al. 2020), while increasing the potential economic value of the study area (Dey et al. 2017; Matthews and Iverson 2017). Beech has shown some signs of resistance to beech bark disease, in genetic (Houston and Houston 2000; Koch et al. 2010; Čalić et al. 2017) and field studies (Leak 2006; Taylor et al. 2013; Cale and McNulty 2018). Additionally, the abundant saplings in our study area are likely relatively less vulnerable than larger trees (Cale et al. 2017). Therefore, beech is likely to persist in our study area, although climate change could compound negative effects of beech bark disease (Stephanson and Coe 2017). The majority of beech bark disease research has been conducted in northern hardwood ecosystems where the invasion has been historically located (Cale et al. 2017), so it is possible that beech bark disease will have different ecosystem effects in our study area than those discussed above (but see: McCann 2016, Rumble et al. 2020).

Additional threats include beech leaf disease, an emerging phenomenon presently found in northern Ohio and adjacent states and Canadian provinces (Ewing et al. 2018). Maples are

also vulnerable to exotic pests; Asian-long horned beetle and spotted lanternfly show potential to decimate multiple *Acer* species (Dodds and Orwig 2011; Dara et al. 2015). Finally, future climate change is predicted to favor many oak and hickory species relative to our focal species (Matthews and Iverson 2017; Iverson et al. 2019b), although red maple demonstrates many traits (Matthews et al. 2011) that may confer ability to respond to climate change (Butler et al. 2015; Iverson et al. 2019b). Oak will likely become less common in mature forests of our study area for at least the next several decades, however, before any potential oak regeneration advantages from climate change will influence the canopy tree composition, judging by regeneration trends shown in this manuscript (Table 5) and oak mortality documented in Radcliffe (2019).

Conclusions

Our ‘taking a step back’ from the predominate focus on oak regeneration may help managers and scientists develop a deeper understanding of conservation problems in topographically complex Central Hardwoods forests. We found that red maple and sugar maple saplings dominate different ends of the soil acidity gradient, that sugar maple saplings are common across slope positions on northeast aspects, and that beech saplings are increasingly common on all but the driest sites. We expect these and other findings to aid recent regional oak restoration planning efforts, by helping managers and scientists identify sites where foregoing oak restoration may result in relatively desirable composition of mesophytic tree species, depending on management goals. It should also assist in planning for the expected invasion of beech bark disease, which is likely to have profound impacts on Appalachian forests in coming decades, and the possible manifestation of other forest health threats. Our findings are from an observational study design, so they should be regarded as suggestive rather than conclusive, and further studies will be needed to determine any causal effects. Additionally, we have focused on

changes in a specific stage of succession, and our results should not be generalized to similar forest types in earlier stages of succession. However, our work is relevant to a common scenario in many present Central Hardwoods forests – an aging oak canopy with a mesophytic-tree dominated understory. Our message is pertinent to many ecosystem types where changing conditions are driving undesired shifts in the forest communities. Individual species matter when planning for biodiversity and ecosystem services on a regional to site level. When resources for restoration are limited, the dynamics of species increasing in dominance over time should be studied with the same level of attention as the species on the decline. Such research could help managers craft more specific plans for targeting multiple conservation goals on a landscape scale.

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543 **Declarations of Interest**

544 The authors declare no conflict of interest.

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Study Area - Athens and Marietta Units of Wayne National Forest, Ohio, USA

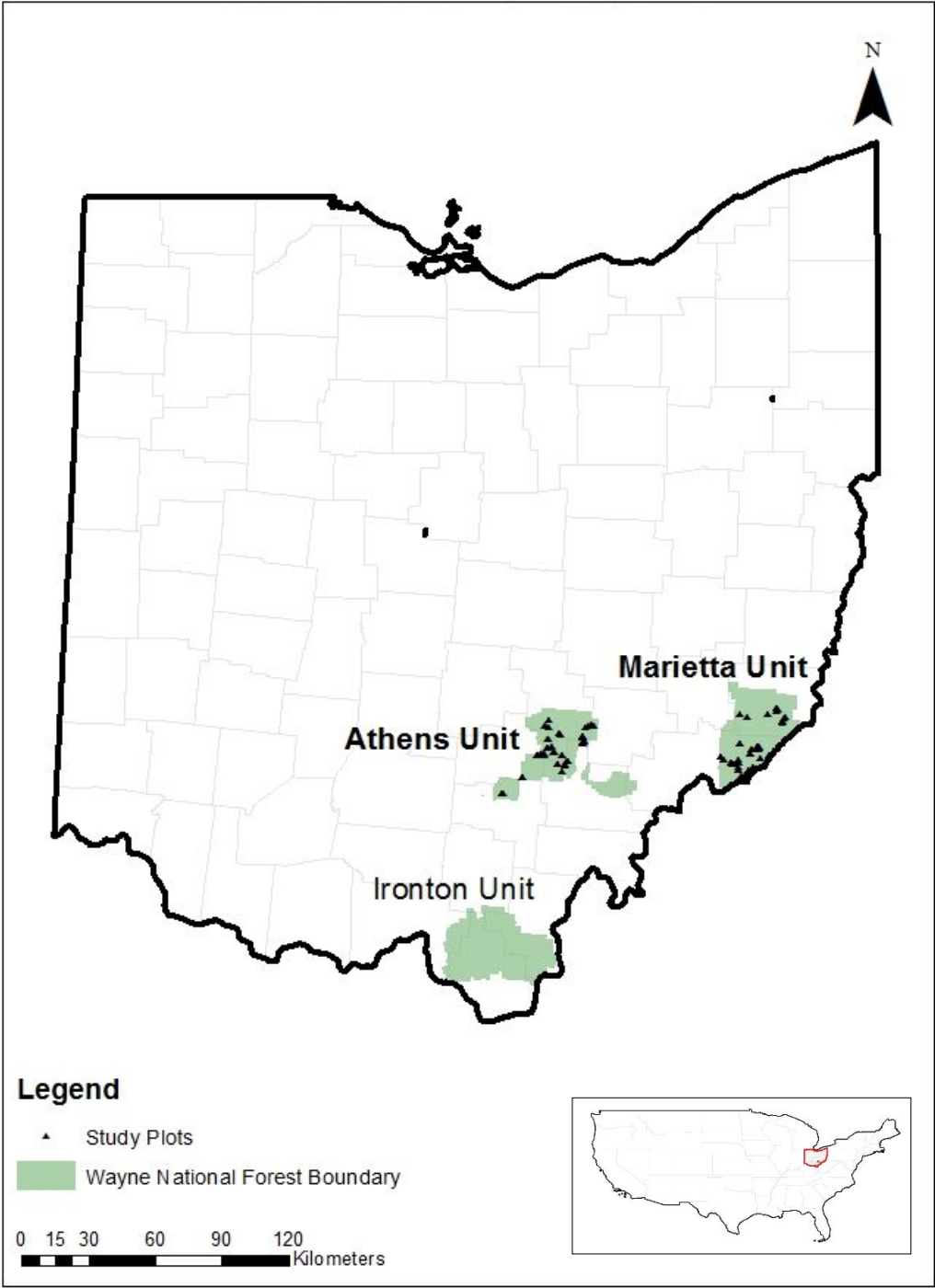


Figure 1: Study area within Ohio. Map produced in ArcGIS Pro 2.5.0 (ESRI 2020). Administrative boundary shapefiles downloaded from publicly available datasets (GEOOhio 2020, US Census Bureau 2020, USDA Forest Service 2020a).

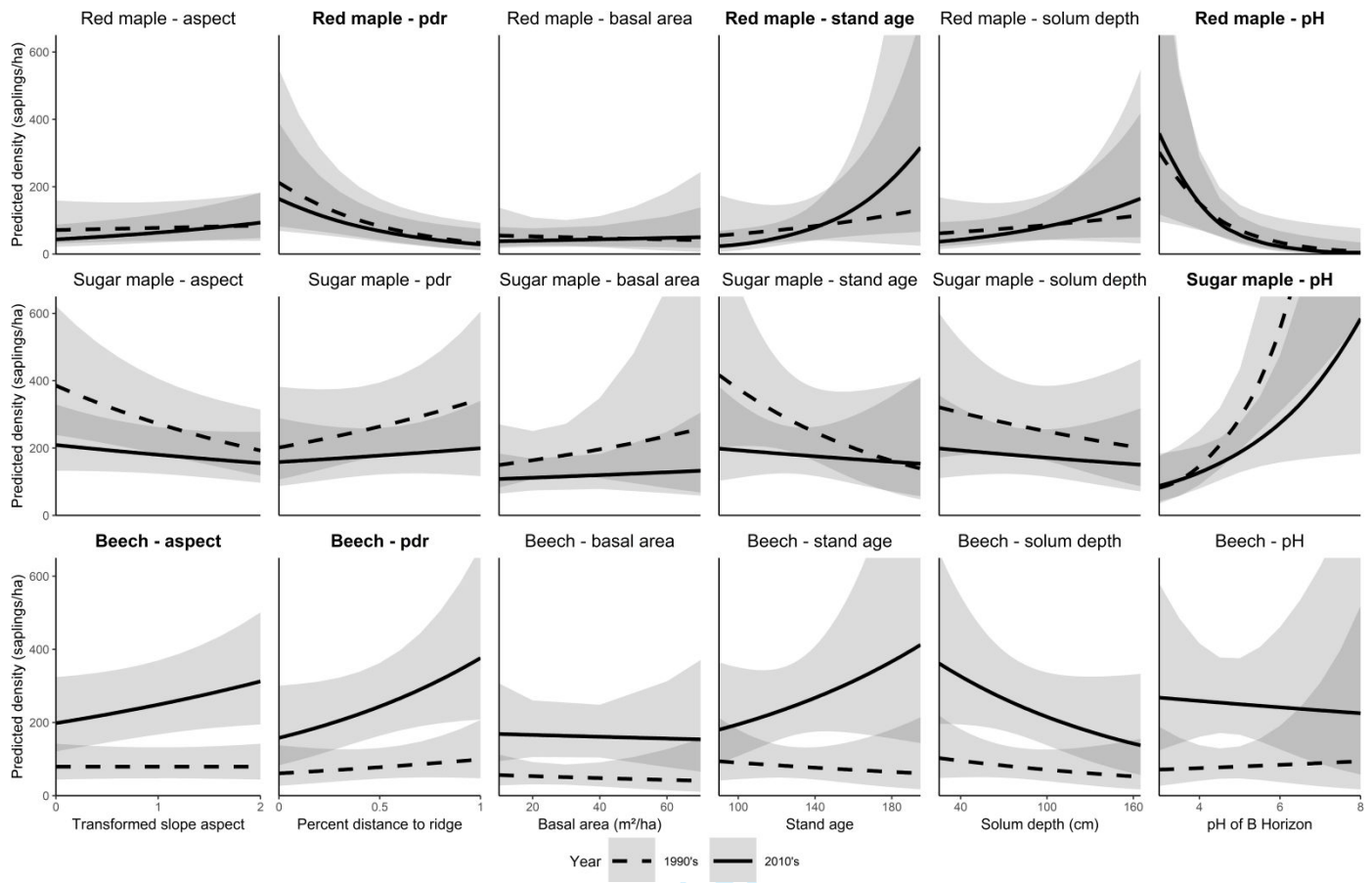


Figure 2: Marginal effects plots of continuous predictor variables in each year, produced using “ggpredict” function in packageggeffects (Lüdtke and Aust 2020). Predictor variables other than that being evaluated are held at their mean values. Bold titles indicate significance in at least one of the two sample periods, and gray ribbons represent 95% confidence intervals.

Table 1: Justification and rationale for predictor variables used in sapling density modeling

Variable	Justification	Studies with importance for our focal species	Notes on units of measure
Plot basal area (m ² /ha)	Used as proxy for subcanopy light in absence of direct measure	Canham 1988 - sugar maple and beech, response to canopy gaps; Poulson and Platt 1996 - sugar maple and beech, response to indirect light indicator	The 1990s models used 1990s basal area, and 2010s models used 2010s basal area. Basal area was calculated from a larger plot (1/20 th ha) than sapling density (1/100 th ha)
Stand age (years)	Important factor in determining successional status and regeneration dynamics of shade tolerant species	Goebel and Hix 1996 - red maple and beech increasing with stand age	Age is stand age, not individual tree age, and was determined from Forest Service Vegetation Management Information System – Hix & Pearcy 1997
Solum depth (centimeters)	Determines area available for root growth, and thus important factor in site productivity	Hix and Pearcy 1997 - thickness of A horizon significant for determining canopy composition	Solum used because fine roots in forests are likely located in upper portions of soil - Schenk and Jackson 2002; late-successional tree species have relatively shallow roots - Gale and Grigal 1987; Pallardy and Rhoads 1993
pH of B horizon	Predicts availability of important plant nutrients, especially calcium	Bigelow and Canham 2002 - all three focal species, saplings, first 10 cm of soil; Arie and Lechowicz 2002 - sugar maple and beech saplings, first 10 cm of soil; St Clair and Lynch 2005 - red maple and sugar maple seedlings, O+A and B horizons.	B horizon used because it was found most significant in study of sugar maple decline - Bailey et al. 2004; and because it is less susceptible to erosion over time than B horizon - Kreznor et al. 1989
Cosine-Transformed Aspect (TASP)	Important determinate of temperature and moisture dynamics	Palus et al. 2018 - all three focal species, multiple life stages, the Wayne; Hix and Pearcy 1997 - predicting overstory composition, the Wayne	Slope aspect was cosine transformed ($\cos(45^\circ - \text{aspect}) + 1$), so it would change continuously rather than breaking between 0° and 360°. TASP maximizes at 2 for a 45° aspect, and minimizes at 0 for a 225° aspect - Beers et al. 1966
Percent distance to ridge (PDR)	Important determinate of temperature and moisture dynamics	Palus et al. 2018 - all three species, multiple life stages, the Wayne; Horsley et al. 2000 - mature sugar maple; Cleavitt et al. 2011 - sugar maple seedlings	Percent distance to ridge is maximized at the bottom of a slope
TASP:PDR interaction	May capture topographic dynamics that TASP or PDR alone cannot	Palus et al. 2018 – beech more common on upper northeast slopes than lower northeast slopes, but more common on lower southwest slopes than upper southwest slopes, the Wayne	High values of the interaction factor indicate upper slopes and northeast facing aspects
Management unit (Marietta)	Will help us understand if unmodeled factors drive sapling-density differences across the two management areas	Seeley et al. 2019 - pre-settlement beech abundance explained by different environmental and climatic variables in Wisconsin and Michigan	The hypothesized relationships are with the Marietta Unit, as that is the additional term in our model

Table 2: Summary statistics of continuous covariates, Athens and Marietta Unit. “Transformed aspect” is cosine-transformed slope aspect ($\cos(45^\circ - \text{aspect} + 1)$), which is maximized at 2 for a 45° azimuth, and minimized at 0 for a 225° azimuth. “PDR” is percent distance to ridge, maximized at 1 in valley bottoms and minimized at 0 on ridgetops. “SE” is standard error

	Transformed aspect		PDR (percent)		Basal area (m ² /ha)		Age (years)		Solum Depth (cm)		pH of B horizon	
	Athens	Marietta	Athens	Marietta	Athens	Marietta	Athens	Marietta	Athens	Marietta	Athens	Marietta
Mean	1.00	1.08	0.53	0.53	29.58	33.38	128.7	135.3	81.7	73.90	4.68	5.08
Median	0.93	1.16	0.55	0.56	29.04	30.68	127.0	130.0	82.0	74.00	4.80	5.00
SE	0.08	0.08	0.03	0.03	0.89	1.46	1.81	1.93	3.69	3.33	0.09	0.06

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Table 3: Mixed effects model coefficients ($\pm$ standard error), p values indicated by asterisks (** $p < 0.01$, * $p < 0.05$, . $p < 0.1$). Coefficients untransformed from model output, generalized linear mixed model with negative binomial distribution assumption and log link

Predictor variable	Red Maple 1990s	Red Maple 2010s	Sugar Maple 1990s	Sugar Maple 2010s	Beech 1990s	Beech 2010s
Intercept	2.952 (2.267)	0.262 (2.185)	-0.487 (1.559)	-0.612 (1.393)	-0.260 (1.796)	-0.700 (1.521)
Transformed aspect (TASP) (0-2)	0.021 (0.401)	0.366 (0.327)	-0.568 (0.294).	-0.397 (0.288)	0.543 (0.355)	0.839 (0.328)*
Percent distance to ridge (PDR) (0-1)	-1.992 (0.972)*	-1.765 (0.851)*	0.106 (0.625)	-0.256 (0.606)	1.542 (0.793).	2.060 (0.694)**
TASP:PDR ((0-2):(0-1))	0.115 (0.652)	0.026 (0.515)	0.416 (0.448)	0.466 (0.438)	-1.016 (0.534).	-1.147 (0.481)*
pH of B horizon	-0.725 (0.315)*	-0.918 (0.318)**	0.652 (0.192)**	0.380 (0.178)*	0.056 (0.251)	-0.035 (0.201)
Depth of solum (cm)	0.004 (0.007)	0.011 (0.006).	-0.003 (0.004)	-0.002 (0.004)	-0.005 (0.006)	-0.007 (0.005)
Stand age (years)	0.008 (0.012)	0.025 (0.012)*	-0.010 (0.008)	-0.004 (0.006)	-0.004 (0.009)	0.008 (0.008)
Basal area of plot (m2/ha)	-0.005 (0.019)	0.004 (0.010)	0.009 (0.010)	0.003 (0.007)	-0.005 (0.014)	-0.002 (0.010)
Unit (Marietta)	0.811 (0.464).	-0.414 (0.452)	0.158 (0.277)	-0.126 (0.261)	0.120 (0.368)	-0.561 (0.298).

Table 4: r^2 values for generalized linear mixed models of sapling abundance, trigamma method. Marginal r^2 value measures variability explained by fixed effects, conditional r^2 value includes both fixed and random effects. The random effect in our model was transect, by which plots were grouped and common soil data was used.

Model	Marginal r^2	Conditional r^2
Sugar Maple 1990s	0.13	0.43
Sugar Maple 2010s	0.04	0.33
Red Maple 1990s	0.15	0.68
Red Maple 2010s	0.27	0.82
American Beech 1990s	0.05	0.68
American Beech 2010s	0.11	0.55

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Table 5: Density (saplings per hectare, “Saps/ha”) and relative density (percent, “Rel. den.”) values for all sapling species, Athens and Marietta Units. These include each individual woody stem 0.1-10 cm dbh, tree or shrub species. Parentheses denote standard error. “Other” row includes all taxa with a relative density value of less than one: *Acer negundo*, *Aesculus flava*, *Aesculus glabra*, *Betula lenta*, *Castanea dentata*, *Cercis canadensis*, *Celtis occidentalis*, *Corylus americana*, *Cornus racemosa*, *Crataegus* spp., *Diospyros virginiana*, *Elaeagnus umbellata*, *Fraxinus quadrangulata*, *Juglans cinerea*, *Kalmia latifolia*, *Liriodendron tulipifera*, *Lonicera japonica*, *Magnolia acuminata*, *Morus rubra*, *Oxydendrum arboreum*, *Pinus strobus*, *Pinus virginiana*, *Prunus americana*, *Prunus serotina*, *Rosa multiflora*, *Rubus* spp., *Sassafras albidum*, *Sambucus canadensis*, *Staphylea trifolia*, *Tilia americana*, *Tsuga canadensis*, *Ulmus americana*, *Vaccinium stamineum*, *Viburnum acerifolium*, *Viburnum dentatum*, *Viburnum* spp., and *Vitis* spp. *Carya* genus includes *Carya cordiformis*, *Carya glabra*, *Carya laciniosa*, *Carya ovata*, *Carya ovalis*, and *Carya tomentosa*. *Quercus* genus includes *Quercus alba*, *Quercus coccinea*., *Quercus montana*, *Quercus rubra*, and *Quercus velutina*. No *Carya* or *Quercus* species had an average relative density value greater than one.

	Athens Unit				Marietta Unit				Combined	
	1990s		2010s		1990s		2010s		1990s & 2010s	
	Saps/ha	Rel. den.	Saps/ha	Rel. den.	Saps/ha	Rel. den.	Saps/ha	Rel. den.	Saps/ha	Rel. den.
<i>Acer saccharum</i>	463 (68)	20	267 (39)	11	532 (69)	20	233 (30)	16	374 (34)	17
<i>Acer rubrum</i>	320 (56)	14	256 (55)	11	467 (88)	17	154 (34)	10	299 (38)	13
<i>Fagus grandifolia</i>	144 (22)	6	457 (74)	19	237 (40)	9	304 (59)	21	285 (31)	13
<i>Cornus florida</i>	326 (58)	14	38 (24)	2	376 (50)	14	6 (4)	0	187 (20)	8
<i>Lindera benzoin</i>	72 (27)	3	206 (46)	9	166 (49)	6	235 (61)	16	170 (29)	8
<i>Ostrya virginiana</i>	241 (43)	10	225 (56)	9	88 (21)	3	48 (12)	3	150 (22)	7
<i>Asimina triloba</i>	49 (39)	2	215 (95)	9	78 (59)	3	29 (15)	2	93 (34)	4
<i>Carpinus caroliniana</i>	102 (25)	4	80 (27)	3	79 (19)	3	54 (19)	4	79 (12)	4
<i>Quercus</i> spp.	72 (15)	3	42 (8)	2	83 (9)	3	50 (6)	3	62 (4)	3
<i>Carya</i> spp.	36 (3)	2	62 (9)	3	74 (10)	3	51 (7)	3	56 (2)	3
<i>Ulmus rubra</i>	52 (16)	2	20 (7)	1	109 (41)	4	7 (3)	0	47 (12)	2
<i>Hamamelis virginiana</i>	4 (4)	0	16 (10)	1	106 (46)	4	54 (25)	4	45 (18)	2
<i>Viburnum prunifolium</i>	96 (59)	4	79 (53)	3	4 (2)	0	1 (1)	0	45 (27)	2
<i>Fraxinus pennsylvanica</i>	49 (18)	2	77 (19)	3	20 (7)	1	23 (8)	2	42 (8)	2
<i>Nyssa sylvatica</i>	25 (7)	1	19 (7)	1	60 (26)	2	20 (8)	1	31 (9)	1
<i>Amelanchier arborea</i>	32 (9)	1	52 (15)	2	24 (9)	1	15 (6)	1	31 (5)	1
<i>Fraxinus americana</i>	10 (5)	0	32 (10)	1	37 (10)	1	44 (20)	3	31 (6)	1
<i>Other</i>	255 (12)	11	259 (10)	11	155 (5)	6	144 (8)	10	203 (2)	9
<i>Total</i>	2348	100	2400	100	2693	100	1471	100	2227	100

Table 6: Density, basal area, and importance value of tree layer, for both the Athens and Marietta Units combined. Parentheses denote standard error. Values reflect stems greater than 10 centimeters diameter at breast height. Importance value is defined as (relative density + relative basal area/2). 'Other' row represents species with an importance value less than one for both sample periods, including: *Fraxinus americana*, *Carya tomentosa*, *Fraxinus pennsylvanica*, *Platanus occidentalis*, *Quercus muehlenbergii*, *Ulmus rubra*, *Carya cordiformis*, *Juglans nigra*, *Tsuga canadensis*, *Prunus serotina*, *Carya laciniata*, *Oxydendron arborea*, *Ulmus americana*, *Ostrya virginiana*, *Aesculus glabra*, *Cornus florida*, *Carya ovalis*, *Sassafras albidum*, *Carpinus caroliniana*, *Pinus virginiana*, *Acer negundo*, *Celtis occidentalis*, *Pinus echinata*, *Populus grandidentata*, *Vitis riparia*, *Prunus avium*, *Cercis canadensis*, *Amelanchier arborea*, *Crataegus* spp., and *Quercus michauxii*.

Species	Density 1990s (trees/ha)	Density 2010s (trees/ha)	Basal Area 1990s (m ² /ha)	Basal Area 2010s (m ² /ha)	Importance Value 1990s	Importance Value 2010s
<i>Quercus alba</i>	55.27 (5.34)	45.70 (4.48)	6.23 (0.58)	7.50 (0.71)	21.92	20.02
<i>Acer saccharum</i>	63.15 (5.40)	69.58 (5.21)	2.04 (0.25)	3.17 (0.34)	15.42	17.64
<i>Quercus montana</i>	25.58 (3.96)	23.64 (3.76)	2.44 (0.38)	3.28 (0.52)	9.26	8.74
<i>Acer rubrum</i>	34.79 (3.57)	39.39 (3.75)	0.92 (0.13)	1.46 (0.18)	7.55	8.73
<i>Fagus grandifolia</i>	21.94 (2.55)	24.24 (2.72)	1.49 (0.25)	1.83 (0.3)	7.78	8.07
<i>Quercus rubra</i>	12.24 (1.53)	11.39 (1.36)	2.24 (0.34)	3.38 (0.49)	6.55	7.20
<i>Liriodendron tulipifera</i>	11.27 (1.93)	11.52 (1.89)	1.69 (0.30)	2.35 (0.40)	5.79	5.87
<i>Quercus velutina</i>	12.73 (1.83)	10.67 (1.48)	1.78 (0.26)	2.38 (0.35)	5.64	5.41
<i>Carya glabra</i>	10.18 (1.75)	8.97 (1.43)	0.44 (0.07)	0.60 (0.1)	2.5	2.47
<i>Quercus coccinea</i>	5.94 (1.54)	4.61 (1.42)	1.00 (0.26)	1.05 (0.31)	2.61	2.13
<i>Carya ovata</i>	6.18 (1.11)	5.70 (1.08)	0.35 (0.07)	0.42 (0.09)	1.72	1.62
<i>Aesculus flavens</i>	3.64 (1.22)	4.97 (1.39)	0.17 (0.07)	0.28 (0.09)	0.77	1.29
<i>Tilia americana</i>	3.52 (0.93)	3.64 (1.02)	0.24 (0.08)	0.31 (0.11)	1.21	1.25
<i>Nyssa sylvatica</i>	4.61 (0.97)	4.85 (1.02)	0.13 (0.03)	0.18 (0.04)	1.16	1.15
Other	37.44 (4.04)	28.95 (3.06)	2.04 (0.28)	2.24 (0.30)	10.12	8.41
Total	308.48	297.82	23.20	30.43	100	100